

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы



Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.



- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

2. Процесс выполнения лабораторной работы



```
irabzalimov@irabzalimov:~$ touch abc1
irabzalimov@irabzalimov:~$ cp abc1 april
irabzalimov@irabzalimov:~$ cp abc1 may
irabzalimov@irabzalimov:~$ mkdir monthly
irabzalimov@irabzalimov:~$ cp april may monthly
irabzalimov@irabzalimov:~$ cp monthly/may monthly/june
irabzalimov@irabzalimov:~$ ls monthly
april  june  may
irabzalimov@irabzalimov:~$ mkdir monthly.00
irabzalimov@irabzalimov:~$ cp -r monthly monthly.00
irabzalimov@irabzalimov:~$ cp -r monthly.00 /tmp
irabzalimov@irabzalimov:~$
```

Рисунок 1: Выполнение примеров



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ mv april july  
irabzalimov@irabzalimov:~$ mv july monthly.00  
irabzalimov@irabzalimov:~$ ls monthly.00  
july  monthly  
irabzalimov@irabzalimov:~$ mv monthly.00 monthly.01  
irabzalimov@irabzalimov:~$ mkdir reports  
irabzalimov@irabzalimov:~$ mv monthly.01 reports  
irabzalimov@irabzalimov:~$ mv reports/monthly.01 reports/monthly  
irabzalimov@irabzalimov:~$
```

Рисунок 2: Выполнение примеров



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ touch may  
irabzalimov@irabzalimov:~$ ls -l may  
-rw-r--r--. 1 irabzalimov irabzalimov 0 Oct  2 18:17 may  
irabzalimov@irabzalimov:~$ chmod u+x may  
irabzalimov@irabzalimov:~$ ls -l may  
-rwxr--r--. 1 irabzalimov irabzalimov 0 Oct  2 18:17 may  
irabzalimov@irabzalimov:~$ chmod u-x may  
irabzalimov@irabzalimov:~$ ls -l may  
-rw-r--r--. 1 irabzalimov irabzalimov 0 Oct  2 18:17 may  
irabzalimov@irabzalimov:~$ chmod g-r,o-r monthly  
irabzalimov@irabzalimov:~$ chmod g+w abc1  
irabzalimov@irabzalimov:~$
```

Рисунок 3: Выполнение примеров



```
irabzalimov@irabzalimov:~$  
irabzalimov@irabzalimov:~$ cp /usr/include/linux/sysinfo.h ~  
irabzalimov@irabzalimov:~$ mv sysinfo.h equipment  
irabzalimov@irabzalimov:~$ mkdir ski.plases  
irabzalimov@irabzalimov:~$ mv equipment ski.plases/  
irabzalimov@irabzalimov:~$ mv ski.plases/equipment ski.plases/equiplist  
irabzalimov@irabzalimov:~$ touch abc1  
irabzalimov@irabzalimov:~$ cp abc1 ski.plases/equiplist2  
irabzalimov@irabzalimov:~$ cd ski.plases/  
irabzalimov@irabzalimov:~/ski.plases$ mkdir equipment  
irabzalimov@irabzalimov:~/ski.plases$ mv equiplist equipment/  
irabzalimov@irabzalimov:~/ski.plases$ mv equiplist2 equipment/  
irabzalimov@irabzalimov:~/ski.plases$ cd  
irabzalimov@irabzalimov:~$ mkdir newdir  
irabzalimov@irabzalimov:~$ mv newdir ski.plases/  
irabzalimov@irabzalimov:~$ mv ski.plases/newdir/ ski.plases/plans  
irabzalimov@irabzalimov:~$
```

Рисунок 4: Работа с каталогами



```
irabzalimov@irabzalimov:~$ mkdir australia play
irabzalimov@irabzalimov:~$ touch my_os feathers
irabzalimov@irabzalimov:~$ chmod 744 australia/
irabzalimov@irabzalimov:~$ chmod 711 play/
irabzalimov@irabzalimov:~$ chmod 544 my_os
irabzalimov@irabzalimov:~$ chmod 664 feathers
irabzalimov@irabzalimov:~$ ls -l
total 0
-rw-rw-r--. 1 irabzalimov irabzalimov  0 Oct  2 18:19 abc1
drwxr--r--. 1 irabzalimov irabzalimov  0 Oct  2 18:20 australia
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Desktop
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Documents
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Downloads
-rw-rw-r--. 1 irabzalimov irabzalimov  0 Oct  2 18:20 feathers
drwxr-xr-x. 1 irabzalimov irabzalimov 74 Oct  2 16:59 git-extended
-rw-r--r--. 1 irabzalimov irabzalimov  0 Oct  2 18:17 may
drwx--x--x. 1 irabzalimov irabzalimov 24 Oct  2 18:15 monthly
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Music
-r-xr--r--. 1 irabzalimov irabzalimov  0 Oct  2 18:20 my_os
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Pictures
drwx--x--x. 1 irabzalimov irabzalimov  0 Oct  2 18:20 play
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Public
drwxr-xr-x. 1 irabzalimov irabzalimov 14 Oct  2 18:17 reports
drwxr-xr-x. 1 irabzalimov irabzalimov 28 Oct  2 18:20 ski.places
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Templates
drwxr-xr-x. 1 irabzalimov irabzalimov  0 Oct  2 16:09 Videos
drwxr-xr-x. 1 irabzalimov irabzalimov 10 Oct  2 16:26 work
irabzalimov@irabzalimov:~$
```



```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
systemd-oom:x:999:999:systemd Userspace OOM Killer:/usr/sbin/nologin
polkitd:x:114:114:User for polkitd:/sbin/nologin
systemd-coredump:x:998:998:systemd Core Dumper:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/usr/sbin/nologin
chrony:x:996:996:chrony system user:/var/lib/chrony:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/usr/sbin/nologin
unbound:x:993:993:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
clevis:x:992:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/usr/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/usr/sbin/nologin
gluster:x:991:991:GlusterFS daemons:/run/gluster:/usr/sbin/nologin
qemu:x:107:107:qemu user:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
openvpn:x:990:990:OpenVPN:/etc/openvpn:/usr/sbin/nologin
```

Рисунок 6: Файл /etc/passwd
Операционные системы



```
irabzalimov@irabzalimov:~$ cp feathers file.old
irabzalimov@irabzalimov:~$ mv file.old play/
irabzalimov@irabzalimov:~$ mkdir fun
irabzalimov@irabzalimov:~$ cp -R play/ fun/
irabzalimov@irabzalimov:~$ mv fun/ play/games
irabzalimov@irabzalimov:~$ chmod u-r feathers
irabzalimov@irabzalimov:~$ cat feathers
cat: feathers: Permission denied
irabzalimov@irabzalimov:~$ cp feathers feathers2
cp: cannot open 'feathers' for reading: Permission denied
irabzalimov@irabzalimov:~$ chmod u+r feathers
irabzalimov@irabzalimov:~$ chmod u-x play/
irabzalimov@irabzalimov:~$ cd play/
bash: cd: play/: Permission denied
irabzalimov@irabzalimov:~$ chmod +x play/
irabzalimov@irabzalimov:~$
```

Рисунок 7: Работа с файлами и правами доступа



```
MOUNT(8)                                System Administration                                MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-fFnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device|mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable]
    mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy,
    rooted at /. These files can be spread out over several devices. The mount command serves
    to attach the filesystem found on some device to the big file tree. Conversely, the
    umount(8) command will detach it again. The filesystem is used to control how data is
    stored on the device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

    mount -t type device dir

Manual page mount(8) line 1 (press h for help or q to quit)
```



```
FSCK(8)                                System Administration                                FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--]
    [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can
    be a device name (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a
    filesystem label or UUID specifier (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or
    LABEL=root). Normally, the fsck program will try to handle filesystems on different
    physical disk drives in parallel to reduce the total amount of time needed to check all of
    them.

    If no filesystems are specified on the command line, and the -A option is not specified,
    fsck will default to checking filesystems in /etc/fstab serially. This is equivalent to
    the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

Manual page fsck(8) line 1 (press h for help or q to quit)
```



```
mkfs(8)                                System Administration      mkfs(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available under Linux. The filesystem-specific builder is searched for via your PATH environment setting only. Please see the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose

Manual page mkfs(8) line 1 (press h for help or q to quit)
```



```
KILL(1)                                User Commands                                KILL(1)
```

NAME

kill - terminate a process

SYNOPSIS

```
kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--]
pid/name...
```

```
kill -l [number|0xsigmask] | -L
```

```
kill -d pid
```

DESCRIPTION

The command **kill** sends the specified signal to the specified processes or process groups.

If no signal is specified, the **TERM** signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the **KILL** signal (number 9), since a process may install a handler for the **TERM** signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a **TERM** signal has been sent, then the **KILL** signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

Most modern shells have a builtin **kill** command, with a usage rather similar to that of the command described here. The **--all**, **--pid**, and **--queue** options, and the possibility to specify processes by command name, are local extensions.

If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS

The list of processes to be signaled can be a mixture of names and PIDs.

Manual page kill(1) line 1 (press h for help or q to quit)

3. Выводы по проделанной работе



В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.